

Lecture 15

Interaction-Oriented Architectures

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- The two major interaction-oriented architectures covered are:
 1. Presentation, Abstraction, Control (PAC).
 2. Model, View, Control (MVC).
 - MVC actually has two closely related variants we will also cover:
 - (a) MVC-I.
 - (b) MVC-II.
- PAC and MVC are closely related.
 - The presentation module is analogous to the view module.
 - The abstraction module is analogous to the model module.
- PAC and MVC are differentiated by their overall structure.
 - MVC is composed of a single model, view, and controller element.
 - PAC is composed of a hierarchy of presentation, abstraction, and control elements.
 - * An ‘agent’ is a group of these three elements in the hierarchy working together.
- PAC agents only communicate with one another by their control elements.
- A new agent is required for each distinct source of data handled by the system.
 - Data sources are distinct if they describe different quantities or are sent to the system at different frequencies.
- There are benefits and drawbacks to using a PAC architecture.
 - Benefits:
 1. Comparatively better at handling multiple, different kinds of data.
 2. Comparatively better at handling data being received at varying frequencies.
 - Drawbacks:
 1. If a PAC hierarchy is deep, the top-level controller takes comparatively long to communicate with the bottom-level controller.
 2. If the PAC hierarchy is shallow, the top-level controller has high coupling.
- An effective implementation of a PAC architecture involves properly balancing the trade-offs associated with different hierarchical organizations of agents.
- MVC is primarily used for simpler, web-based systems.
 - MVC emphasizes the system’s look-and-feel.
- Certain software principles may be occasionally ignored if violating it is more advantageous than adhering.
- MVC-II is the usual variant of MVC discussed above.

- MVC-I is a simpler variant of MVC composed of two elements:
 1. A model module.
 - Forms the system's 'back-end'.
 2. A single view/control module.
 - Forms the system's 'front-end'.
- MVC-I violates the separation of concerns principle.
 - For simple systems with a minimal need for a separate controller module, this is worth it.
- The subscribe/notify pattern is used to implement communication between elements in MVC modules.
 - In the subscribe/notify patterns, elements can be either:
 1. Subjects.
 2. Observers.
 - Observers can subscribe to subjects to be notified of state changes.
 - Observers can unsubscribe to subjects to stop receiving such notifications.
 - Subjects track all subscribed observers.
 - Subjects have a special method to notify all subscribed observers.
- In the MVC-I architecture:
 - The back-end is the subject.
 - The front-end is the observer.
- In the MVC-II architecture:
 - The model element is the subject.
 - The controller and view elements are both observers of the model element.